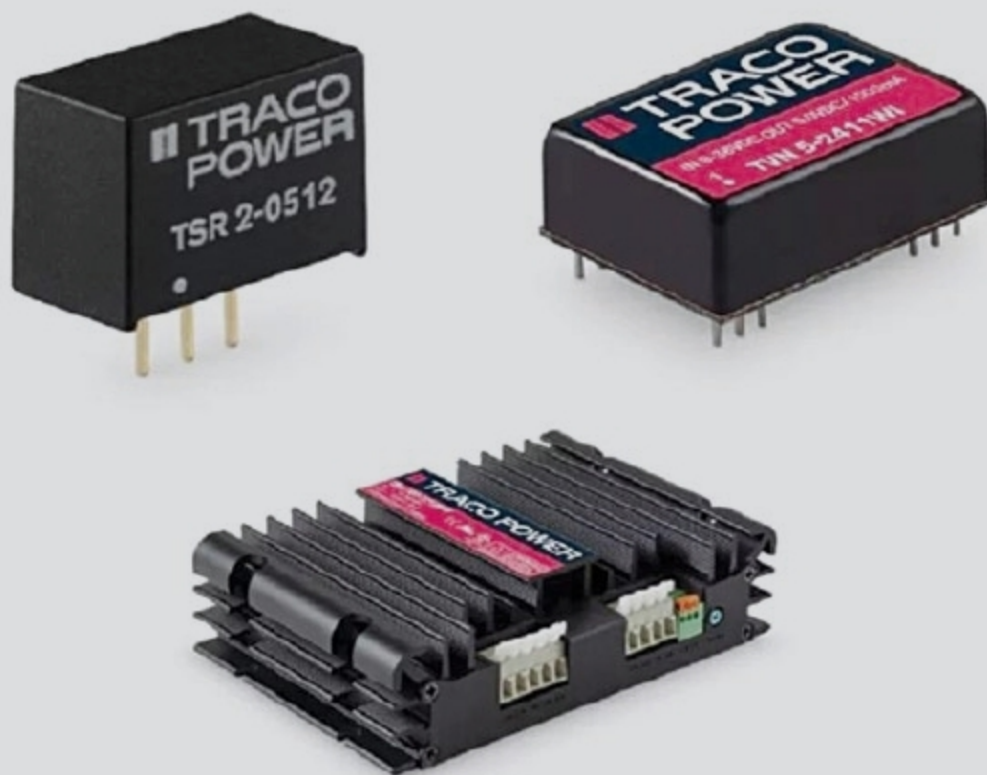


How To Choose The RIGHT POWER SUPPLY For Your VEHICLE

One of the topics that car owners love to talk about time and again is their vehicle's performance data. This article provides a guide on how to choose the right power supply for your design, starting with accurate DC/DC converter measurements



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Many people have already said that consumers are being misled when their cars consume more fuel than the manufacturer data specifies. But is misled the right term to use? Or are we dealing with a subject that designers in the electronics industry have long been familiar with, that data collected in the laboratory differs from data collected from actual usage?

How does that happen? Are readings actually incorrect, or are there considerations that need to be taken into account the moment an AC/DC power supply or a DC/DC converter is integrated into a circuit?

Ultimately, it should be clear to everyone that a car tested with no weight inside on a defined test track consumes less than

when it is fully-loaded with a family of four on the side of the nearest mountain.

As a rule, personal circumstances, such as budget, will determine which car we buy. But how do we go about choosing the right components to deliver power?

Analysing the requirement

The first thing we should do is define what the respective power supply is required for. This throws up some apparently simple preliminary questions. The key objective is to bring input voltage for the application to a new potential.

- Should potential be divided or not?
- What are the input and output voltage ranges, and what is required at which output current?